Chapter 2

Exoplanets, a primer

This thesis sits right at the interface of two different disciplines: Machine learning (i.e., computer science) on the one hand, and exoplanet science (i.e., astrophysics) on the other. To appreciate its contents, a certain understanding of both sides is useful. The main parts in chapters 3, 4, and 5 are written in a way that assumes that the reader is familiar with the astrophysical context, and focuses more on explaining the methodical innovations that are being introduced. To make things more accessible to people without a strong background in astronomy, this chapter offers a first introduction to the field of exoplanet science. Its goal is to provide an overview of selected concepts and ideas that set the stage on which the main contributions of this thesis take place. It can also be seen as an (incomplete) summary of things I have learned during the course of my Ph.D. studies and have found useful in my interactions with exoplanet scientists. Of course, given the vastness of the subject, this introduction can only scratch the surface and will be by no means complete or comprehensive. For this reason, an extensive bibliography is provided, which can serve as a starting point for the interested reader. Moreover, I advert to the books by Seager (2010b), Perryman (2018), and Deeg & Belmonte (2018), who provide in-depth reviews of virtually every aspect of exoplanet science.

Contributions
I have written and edited this chapter based on my understanding of the subject and my review of the literature. Unless explicitly noted otherwise, all figures have been created by me.

***Advice for reading this chapter**. This chapter is written such that the individual sections at least partially build on each other. Nevertheless, if a reader finds themselves pressed for time, I recommend focussing on sections 2.4 (High-contrast imaging) and 2.5.2 (Atmospheric retrieval), which are most directly relevant to the main part of this thesis.*

2.1 WHAT IS AN EXOPLANET, ANYWAY?

"I work on machine learning for astrophysics, specifically exoplanets. Um, do you know what an exoplanet is?" —— This exact sequence of words is how many conversations have started for me over the past few years. More often than not, the answer was but a slightly abashed smile, which has led me to believe that perhaps it would only be appropriate to